TEM Workbench code review

Nov 3, 2026 · @Someone

Scope and limits

Reviewed commit: `166cab9` on `main` (3 November 2026, "Add batch map PNG choices and independent centre spacing"). The brief was prepared at `cb4ca63`; the three later commits change overlap/batch code and the default FFT minimum separation (0.12 → 0.50 nm^{-1}). That separation change matters for the detector findings below. All 152 tests pass at `166cab9` (2 skipped) on Python 3.12.3, NumPy 2.4.4, SciPy 1.17.1.

This was an audit only. No repository files were changed. All experiments ran on copies of the engine outside the app.

Data limits, stated up front:

- No real images were available. The DM4 demo packs (ST1221, ST1144, ST1203) are hosted on Dropbox, which this sandbox cannot reach, and ST1156 and its diagnostic exports are not in Git. Every measurement here uses synthetic lattice images with known answers. ST1156 statements are taken from `docs/implementation-plan.md`, not reproduced.
- The sandbox has 1 CPU. Single-region timings are real; multi-worker grid throughput could not be measured. The 3.1× four-worker figure in `docs/parallel-performance.md` is the only grid scaling evidence.
- Synthetic images are idealized (sinusoidal lattice fringes, Gaussian noise, optional amorphous background). They test the code path, not HRTEM contrast transfer. Treat accuracy numbers as code-behaviour checks, not as validation of phase identification.

Summary

Three implementation defects explain most of the weak-pattern and crop-size behaviour seen in synthetic tests. All three have narrow, testable fixes.

1. **Second orientations are never searched.** `scale_fitting._seed_candidates` keeps the top 3 hypotheses per c/a ratio *before* removing near-duplicates, so the slots fill with copies of the dominant pattern. A second HCP orientation carrying 47% of the peak weight was never refined. Suppressing duplicates first recovered it at 0.6× and 0.35× amplitude with no change to five single-crystal controls.

2. **Rank-auto is biased toward the smallest peak set.** Its score is normalized by each trial's own peak weight, so dropping unexplained peaks always helps. It chose 4–7 peaks almost everywhere and removed an equal-strength second orientation that manual and geometry-auto both reported.
3. **The detector cutoff follows the strongest peak.** The 0.35 × max rule rose from 1.95 to 3.66 between 4 and 30 nm crops while noise maxima stayed at 2.0–2.6. Genuine reflections above every noise maximum were excluded, and auto modes cannot recover them because they only raise the cutoff.

Performance is dominated by rank-auto repeating full crystal matching at every cutoff: about 0.19 s per level, so 12.4 s for 65 levels versus 0.27 s manual. A results-preserving cache of weak-peak preparation saves 9–32%. The larger gains need the search fixes and a cheaper outer objective, which change results and must be validated.

Nothing here establishes phase identities. Synthetic images test code behaviour only.

Prioritized findings

Type: **Defect** = demonstrable implementation error; **Assumption** = scientific choice needing validation; **Perf** = speed only.

#	Type	Finding	Location	Evidence
1	Defect	Hypotheses are truncated to the top 3 per c/a ratio before near-duplicates are removed; final dedup treats the same angle with a different scale as distinct. A second orientation is never refined.	`scale_fitting._seed_candidates`	HCP zone [010] at 23° + 73° (0.6×): 6 second-pattern profiles, ~47% of weight. All 3 seeds were at 23° (scales 1.000/1.008/0.984). Dedup-first patch: `multiple_orientations` at 0.6× and 0.35×, 12 and 20 nm.

#	Type	Finding	Location	Evidence
2	Defect (likely)	Same pattern in fixed-scale mode: top 2 angles per template at 0.5° rounding, so 23.0° and 23.5° can fill both slots.	`analysis_core.search`	Code reading; not yet tested with the patch.
3	Defect	Rank-auto's objective rewards shrinking the peak set: coverage is divided by the trial's own total weight.	`engine.analyze`, rank branch; `core.score_hypothesis`	Kept fewer peaks than manual in 21 of 30 two-pattern runs (4–7 peaks in all 30). Equal-amplitude second HCP orientation: manual/geometry `multiple_orientations`, rank `hcp_compatible`. HCP + BCC: rank removed every BCC spot even at 1.0×.
4	Defect	Initial cutoff = max(1.8, 0.35 × strongest enrichment). It scales with crop size; noise does not. Auto modes only raise it.	`analysis_core.spectrum`	HCP [0001]: cutoff 1.95 → 3.66 from 4 to 30 nm; noise maxima 2.0–2.6. At 30 nm, second-order spots at 3.2 (above every noise maximum) were excluded.
5	Defect	Weak-maximum detection hard-codes smoothing 0.65 bins and level 1.8, ignoring user detector settings.	`peak_evidence.prepare`	Code reading. With smoothing set to 1.3, detection and weak-peak evidence use different signals.

#	Type	Finding	Location	Evidence
6	Assumption	Four matched profiles are required. First-order HCP [0001] and BCC [111] give only 3 unique profiles, so they need second-order spots. At small crops, auto modes reach four by admitting noise.	`core.score_hypothesis` (`len(matches)>=4`), `classify`	HCP [0001]: `insufficient` at 6–8 nm. BCC [111]: `insufficient` at 6–20 nm, and labelled `hcp_compatible` at 4 nm from 4–5 spurious peaks.
7	Assumption	Coverage and validity use the whole peak set as denominator. A real weaker pattern is charged for not explaining the dominant one.	`core.score_hypothesis`, `core.classify` (`u>.15`)	Follows from the formula; contributes to findings 1 and 3.
8	Assumption	Zone library is unbalanced: HCP 18 zones, FCC 13, BCC 5 (no [012], [112]). The chance penalty is per template and does not account for how many templates were tried.	`templates.json`, `crystal_templates.fcc_templates`	Counted from the library. More tries per phase means more chances to fit noise.

#	Type	Finding	Location	Evidence
9	Assumption	HCP gets 41 c/a seed ratios and an extra fitted parameter; cubic gets one. The scale cost (0.10 per 5%) is uncalibrated against that extra flexibility.	`scale_fitting.search`, `decomposed_cost`	Code reading. Not yet measured with matched noise-only controls.
10	Defect (minor)	Duplicate removal differs: fixed mode merges same-phase candidates across zones, auto modes only within a zone.	`core.retain` vs `scale_fitting.retain`	Code reading.
11	Assumption	Ring background uses the mean of log power, which includes the peaks themselves.	`analysis_core.spectrum`	Median removed 2–3 false maxima above the cutoff at 4 nm; small effect at ≥6 nm.
12	Assumption	Smoothing in FFT bins suits window-limited peaks; heavier smoothing hurt here.	`analysis_core.spectrum`	12 nm, second order: true/noise 2.9/2.41 at 0.65 bins, 1.6/1.62 at 1.3 bins. At 0 bins, 6–11 noise peaks passed at 4 nm. Broadened-peak results are in the validation section.

#	Type	Finding	Location	Evidence
13	Perf	Weak-peak preparation is recomputed for every cutoff, though only the strong-peak exclusion depends on it.	`peak_evidence.prepare` via `engine.analyze`	Cached prototype: identical output; 2.07 → 1.40 s and 7.99 → 7.28 s.
14	Perf	Geometry-auto recomputes `core.purity` on the already-selected set.	`engine.analyze`	Code reading; cost is small (~7 ms per call).

Findings 1–4 interact. Finding 4 removes weak true spots before matching, finding 1 stops the matcher seeing a second pattern among what remains, and finding 3 then prunes it. Fixing only one of them will not make weak second patterns reliable.

Performance breakdown

Rank-auto costs about 0.19 s per tested cutoff in HCP overall + c/a mode, so its runtime grows with the number of distinct peak strengths. Manual and geometry-auto stay under 1 s for titanium settings. Unknown material is slow in every mode because of the orthorhombic search.

Median of three unprofiled runs, one region, 1 CPU, synthetic HCP [010] + weak BCC [011] image:

Case (crop, levels tested)	Manual	Geometry-auto	Rank-auto
Fixed scale, 20 nm, 6 levels	0.08 s	0.09 s	0.51 s
Fixed scale, 20 nm, 65 levels	0.13 s	0.61 s	2.89 s
HCP overall + c/a, 5 nm, 4 levels	0.36 s	0.46 s	1.40 s
HCP overall + c/a, 20 nm, 6 levels	0.22 s	0.23 s	1.52 s
HCP overall + c/a, 20 nm, 65 levels	0.27 s	0.74 s	12.4 s

Case (crop, levels tested)	Manual	Geometry-auto	Rank-auto
Unknown material (all four phases), 20 nm, 5 levels (profiled)	4.2 s	4.1 s	18.7 s

The 65-level rows lower the relative cutoff (`threshold` 0.2 instead of 0.35) to create a realistic large peak pool; the app caps pools at 70 profiles.

Where rank-auto time goes, 65-level HCP overall + c/a case (cProfile): hypothesis seeding 56%, least-squares refinement 19%, weak-peak preparation 15%, peak-evidence inspection with 11 rotated controls 10%. FFT and geometry screening are each under 1%. In Unknown mode the orthorhombic fit is about 60% of every mode's time.

What can be reused across nested peak sets

- **Preserves results:** weak-peak candidates (log residual, local maxima, annulus statistics) are independent of the trial; only the strong-peak exclusion changes. Prototype gave identical JSON output and saved 0.7 s at 65 levels.
- **Preserves results:** `prediction_points(template, 1, ratio)` and the pair hypotheses from the 16 strongest peaks are identical for every trial with ≥16 peaks. Caching them per crop is exact; nearest-neighbour support would need a radius query over the full pool, filtered per trial, to stay exact.
- **Preserves results:** geometry-auto can reuse the selected trial's `purity` result for `periodicity`.
- **Changes the search:** inspecting peak evidence only for candidates that could still be retained, warm-starting refinement from the previous level, or testing fewer levels.

Grid throughput

Not measurable here (1 CPU). Per-region results are 11–71 KB pickled, so queue transfer is unlikely to matter. Numeric-library and FFT threads are already pinned to 1 per worker. The 3.1× four-worker speedup in `docs/parallel-performance.md` is consistent with CPU-bound regions and is the only scaling evidence available.

Validation matrix

All three modes are reliable on clear single crystals with at least four visible profiles and on pure noise. They fail on weak or equal second patterns and on 3-profile hexagonal patterns. Synthetic images: 0.02 nm pixels, Gaussian noise 1.0, amorphous background 0.5, titanium references, HCP overall + c/a mode unless noted.

Scenario	Crops	Manual	Geometry-auto	Rank-auto	Note
Single HCP [010], 6+ profiles	4–30 nm	Correct at 6–30 nm; `insufficient` at 4 nm	Correct at all sizes	Correct at all sizes, but keeps only 4 of 6 peaks	Positional agreement 1.0 whenever labelled
Single BCC [001]	4–30 nm	Correct	Correct	Correct	Stable across crop size
HCP [0001], 3 first-order profiles	4–30 nm	`insufficient` at 4–8 nm	Correct at 4 nm via 4 spurious peaks; `insufficient` 6–8 nm	Same as geometry	Needs second-order spots; finding 6
BCC [111], 3 first-order profiles	4–30 nm	`insufficient` to 20 nm	**`hcp_compatible` at 4 nm**	**`hcp_compatible` at 4 nm**	Wrong phase from spurious peaks
HCP [010] + BCC [011], BCC at 1.0/0.6/0.35×	12, 20 nm	BCC never accepted; 1–2 of 5 BCC profiles detected	Same	All BCC spots removed	Detector cutoff and BCC [011] weak higher orders limit this
HCP [010] at 23° + 73°, equal amplitude	12, 20 nm	`multiple_orientations`	`multiple_orientations`	`hcp_compatible`	Rank bias; finding 3
HCP [010] at 23° + 73°, 0.6× and 0.35×	12, 20 nm	`hcp_compatible`; second pattern never explored	Same	Same	Seed truncation; dedup-first patch fixes manual
HCP [0001] two rotations	12, 20 nm	`unresolved` at 1.0×	`hcp_compatible`, one pattern	Same	Each pattern has only 3 profiles
Noise only (3 seeds, 2 positions)	4–20 nm	`insufficient`, 0 peaks	`unresolved`; admits up to 70 noise peaks at 20 nm	`unresolved`	No false crystal labels in 90 runs
Crop-size, HCP [0001]	4–30 nm	Cutoff 1.95 → 3.66	Same pool	Same pool	Genuine second-order spots excluded at 20–30 nm

Two method checks:

- **Coarse-to-fine cutoff search** (every 3rd or 6th level, then refine around the best): matched rank-auto's full search in 30 of 30 simulations while evaluating 40–60% of levels. It missed geometry-auto's selection in 7 of 30. Rank agrees mainly because its objective drifts toward the smallest set; that is not evidence the shortcut is safe after finding 3 is fixed.

- **Intrinsically broadened peaks** (smooth random phase, 2 and 5 nm correlation length): every smoothing option detected the same 6 strongest profiles with no spurious peaks. The test was too mild to separate bin-based from constant reciprocal smoothing; it needs stronger broadening or real data.

Still needed on real data, treating stored labels as hypotheses: ST1156 at 4–30 nm (the crop-size case), ST1203 (strong BCC candidate), ST1221, and at least one region a user believes shows two orientations of one phase.

Immediate fixes versus experimental alternatives

The immediate fixes are either exact speedups or corrections to clear defects. Everything that changes what counts as a peak or a pattern stays behind an option until checked on real data.

Immediate

Change	Results change?	Expected benefit	Risk
Cache weak-peak candidates per crop; filter by trial peaks	No (identical JSON in 4 tests)	9–32% faster rank-auto	Cache key must include smoothing, calibration and crop
Cache `prediction_points` per template/ratio and reuse selected `purity`	No	Small, mostly in rank-auto seeding	Low
Remove near-duplicate hypotheses (angle within ~2°, scale within 0.02) before keeping the top seeds; require distinct orientations in the final seed set	Yes, adds candidates	Second orientation recovered at 0.6× and 0.35×; single crystals unchanged in 5 controls; slightly faster	More candidates may raise false secondaries; check noise and harmonic cases
Apply the same fix to the top-2 angles in `core.search`	Yes	Fixed-scale mode gets the same recovery	Untested

Change	Results change?	Expected benefit	Risk
Pass user smoothing and floor into `peak_evidence.prepare`, or record that it uses fixed values	Yes when settings differ from defaults	Detection and weak evidence use the same signal	Saved results with non-default smoothing change on rerun
Report a flag when fewer than 4 profiles are detected in a hexagonal 3-profile pattern	No	Users see why HCP [0001] / BCC [111] stay `insufficient`	None
Align duplicate removal between `core.retain` and `scale_fitting.retain`	Yes, minor	Same retention rule in every mode	Low

Experimental (behind an option, validate first)

Alternative	Suited to	Why	Main risk
Noise-referenced cutoff: robust spread of the detector signal in the band, set for a target false-peak rate	TEM FFT	Noise maxima were flat at 2.0–2.6 across crop sizes while the strongest-peak rule rose to 3.66	Real FFTs have non-white backgrounds; needs per-image checks
Ring median background	TEM FFT	Removes self-suppression from peaks in their own ring; helped at 4 nm	Small gain at ≥6 nm
Smoothing width = combination of window width (bins) and an intrinsic width (nm^{-1})	TEM FFT	Window broadening is fixed in bins; specimen broadening is fixed in nm^{-1}	My broadening test was inconclusive
Local prominence against an annulus, as already done for weak maxima, applied to primary detection	TEM FFT and diffraction	Consistent with existing weak-peak code; less dependent on global maximum	Shoulders of strong peaks

Alternative	Suited to	Why	Main risk
Subpixel Gaussian fit on the background-subtracted log signal, with a per-peak position uncertainty	Both	Matching radius could reflect actual uncertainty instead of window width alone	Fit failures on merged peaks
Watershed separation of merged components	Mainly direct diffraction	Real-space FFT peaks at these crop sizes are mostly window-limited and separated by the 0.50 nm^{-1} rule	Over-splitting noisy FFT peaks
Background-subtracted integrated intensity	Direct diffraction	Physically meaningful there; FFT intensity depends on contrast transfer and thickness	Over-interpreting FFT power
Pattern-specific coverage for the second pattern (explained weight / weight in that pattern's predicted neighbourhood)	Both	Global denominator charges a real weak pattern for the dominant one	Encourages fitting noise without a chance correction
Pair objective: union of explained weight, shared profiles counted once, chance penalty per pattern, one complexity cost per extra pattern and per extra scale parameter	Both	Replaces both current auto objectives with one rule	Weights need calibration on noise-only and single-crystal controls
Look-elsewhere correction for the number of zones tried per phase	Both	HCP has 18 zones vs BCC 5	Changes rank comparisons between phases

Coarse-to-fine cutoff search is not recommended yet. It matched rank-auto in 30 of 30 simulations, but only because rank-auto drifts to the smallest set; it missed geometry-auto's choice in 7 of 30.

Implementation sequence, benefits and risks

Start with tests that pin the defects, then exact speedups, then one search fix at a time. Each step should leave existing saved results reproducible under their recorded engine version.

1. **Add a synthetic regression suite.** Cases from the validation matrix: single HCP [010] and BCC [001], two HCP [010] orientations at 1.0/0.6/0.35×, HCP + BCC, HCP [0001] and BCC [111] at 4 nm, and noise-only crops. Assert labels and that the second pattern is *explored* (present in the candidate pool), not just accepted. Gate: current code fails the two-orientation cases, as shown here.
2. **Exact speedups.** Cache weak-peak candidates per crop, cache prediction points per template/ratio, reuse the selected `purity`. Gate: byte-identical result JSON on the suite and on ST1203/ST1221 crops. Benefit: 9–32% on rank-auto. Risk: stale caches if keys omit a setting.
3. **Seed deduplication fix** in `scale_fitting._seed_candidates`, then `core.search`. Gate: suite passes; noise-only crops still give no crystal labels; ST1156 crop-size results recorded before and after. Benefit: second orientations become visible to classification. Risk: more marginal secondaries, so watch `multiple_orientations` rates on real single-crystal regions.
4. **Settings consistency** in `peak_evidence.prepare` and retention rules. Gate: unchanged at default settings.
5. **Experimental detector option:** noise-referenced cutoff, optionally with ring median. Run the existing crop-size diagnostic on ST1156 with old and new detectors side by side. Benefit: fewer genuine spots lost at large crops. Risk: more noise peaks where the FFT background is not white.
6. **Experimental pair objective** replacing global-denominator coverage for secondaries, with explicit shared-profile, chance and complexity terms. Calibrate the terms on noise-only and single-crystal controls before comparing on mixed regions. Only then revisit rank-auto: either drop its per-trial normalization or keep it labelled as "dominant pattern only".
7. **Performance follow-ups** once results are stable: candidate pruning before peak-evidence inspection, warm-started refinement across levels, and a re-test of coarse-to-fine search under the new objective.

Not recommended: summing top-two ranks, tuning penalty weights to raise pass rates, or switching smoothing defaults before step 5's real-data comparison.

The review scripts used here (synthetic generator, profiling, cache and seed patches) are in the sandbox, not the repository. I can turn steps 1–2 into a branch on request.